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Social Entrepreneurship: Innovations for Sustainable Development

# Novel data-driven model for flood prediction using big data: Social resilience and risk management

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**Abstract** Among the most destructive natural catastrophes, floods can cause significant disruptions that have long-term effects on the environment, society, and economy. Reducing hazards and improving readiness require efficient flood prediction models. To predict flood events, traditional flood forecasting methods mostly use hydrological and meteorological data. However, these approaches, frequently overlook the larger socioeconomic elements that affect community readiness and flood resistance. To overcome this limitation, a new data-driven method has been presented that combines big data analytics with social resilience and risk management frameworks. This model integrates important socioeconomic variables with large datasets to improve flood prediction accuracy, and the data is subjected to a rigorous preprocessing step that includes cleaning and Z score normalization to ensure the quality, consistency, and reliability of input variables. Principal Component Analysis (PCA), a statistical method that preserves the most significant variation while reducing the dimensionality of data, is used to extract features. PCA improves model performance and computing efficiency by streamlining the dataset without sacrificing important information. The Bison Optimized Efficient Deep Neural Network (BO-EDNN) is used in the research to anticipate floods. In contrast to traditional deep learning models, BO-EDNN dynamically adjusts to intricate flood patterns, enabling more accurate predictions. Conventional flood prediction models ignore the socioeconomic factors that influence disaster response and recovery in favor of concentrating only on physical dangers. According to empirical findings, the BO-EDNN model performs noticeably better than conventional techniques. 95.2% prediction accuracy, 96.1% precision, 92.6% recall, and 93.2% F1 score are attained. The robustness and dependability of the model are demonstrated by these excellent metrics. The incorporation of social resilience components, such as infrastructure vulnerability, adaptive capability, and community readiness, is a significant innovation of this approach. Conventional flood prediction models ignore the socioeconomic factors that influence disaster response and recovery in favor of concentrating only on physical dangers. The robustness and dependability of the model are demonstrated by these excellent metrics.

**Keywords:** flood prediction, social resilience, risk management, social resilience, socio-economic factors, line spacing

## 1. Introduction

Floods are the most dangerous since they cause many deaths, financial losses, and unstable economies globally. Preparation and flood control are essential for minimizing this negative impact and creating a flood-resilient community (Kommanaboina, 2024). One of the most common and upsetting natural disasters in many metropolitan areas is flooding, particularly in light of the effects of urbanization and climate change. Forecasting flooding is essential for assessing and managing urban floods and can offer useful tools for making decisions that lessen the impact of flooding on the environment and society (Zhou et al., 2023). Nevertheless, quantifying flood risk is necessary to effectively create improved prevention plans and ultimately improve resilience. Flood risk is defined as the anticipated harm that arises from the likelihood that a risk will occur in conjunction with the vulnerability and exposure of the at-risk community while taking into account various uncertainties (Abdel-Mooty et al.,2022). Unless the possible advantages are precisely stated and measured, it is frequently challenging to invest in flood adaptation measures due to conflicting objectives in resource-constrained developing nations (Narender et al., 2024).

Improving resilience is essential to attaining sustainability, especially in a nation vulnerable to extreme natural disasters such as earthquakes and floods, which have historically resulted in tremendous financial losses and a high death toll

because of geographical and ecological vulnerabilities. According to one definition, resilience is the ability of an intricate system to recover after an environmental or human-created disaster (Pour et al., 2024). For real-time flood forecasting, hourly rainfall forecasting and estimation are crucial. Therefore, the most important component of the early warning system during catastrophic monsoon occurrence is the accurate and dependable hourly flood prediction approach. The floodplain's residents receive timely warnings from actual flood forecasts, warnings, and response systems (Hussain et al., 2021). While crisis management stresses insights such as evaluating population protection measures, flood repercussions, limited access to essential facilities, and spatial variations in recovery, predictive monitoring of flooding can reveal near-future dangers and economic disruptions (Yuan et al., 2022). The severe effects of floods and the increasing severity of the danger of flooding in cities highlight how crucial it is to characterize flood risk throughout cities to identify areas most at risk from flooding and examine the fundamental social factors that lead to locations in greater danger of flooding. To achieve flood risk reduction, integrated urban planning requires knowledge of the geographic distribution of flood risk in cities and the identification of high-risk locations (Yin & Mostafavi, 2023).

To increase community resilience, Nakhaei et al. (2024) attempted to develop a creative structure for holistic risk management in areas of drought, with a focus on risk assessment, making choices, and flood prediction. Techniques include statistical outlier trimming, artificial intelligence-based forecasts support vector machines, artificial neural networks, neighbor classification (SVMs, ANNs, NNCs), flash flood modeling via a watershed modeling system, geographic information system (GIS) evaluations, and Delphi decision-making. High $R^2$ scores, as well as high flood risk identification and sustainable remedies, were provided by the results. Dong et al. (2022) improved flood forecast accuracy by combining physical sensor data and sociological reports via a multivariate time series technique. The model forecasted 24-hour flood risks on the basis of Harris County and TX data and 7 historical flood events. The results revealed that the integrated dataset outperforms standalone data, with 82.5% accuracy. Lin et al. (2023) created a near-action-time risk flood assessment system for data-scarce situations via social sensors and a region-stable deep neural network (RS-DNN). The method allows for rapid, large-scale flood risk assessments using publicly available data by integrating social sensing for catastrophe information and risk distribution factors. The results show adaptability between cities and disaster procedures, resulting in efficient, portable mapping.

Ghaith et al. (2022) proposed a fast, data-driven flood prediction approach (FPM) that combines stochastic incident synchronization, the convolutional neural network-long short-term memory (CNN-LSTM) method, and statistical model averaging. It accurately predicted the extent of flooding and depth, with >95% accuracy and >80% depth, while being 99.5% quicker than hydraulic modeling. It has been demonstrated in Calgary and could help with early warning, city development, and climate resilience. Hussain et al. (2023) aimed to analyze flood resilience in that nation via an integrated flood vulnerability model that combines machine learning (ML), GIS, remote sensing, and the analytical hierarchy process (AHP). Random forest and SVM provide flood susceptibility maps, whereas AHP computes capacity scores. These findings suggest moderate to high resilience in the northwestern and southwestern regions but minimal resilience in the central and southeastern regions. Chen et al. (2021) used 6ML models, which included the CNN, XGBoost, and variant boosting decision tree (GBDT), to improve flood risk assessment in the Pearl River Delta. With hyperparameter optimization for model fairness, it trains using 2,000 sample points and 12 indices. The findings indicated that the GBDT model was the best and that road density, the digital elevation model (DEM), and maximum 1-day precipitation (M1DP) all had major impacts on flood risk, providing information for flood management.

A real-time decision support system (DSS) for flood catastrophe management in the nation was designed and implemented by Akbarian et al. (2022) by combining multicriteria decision-making (MCDM), machine learning (ML), and hydrological data mining. While other machine learning techniques effectively forecast floods, K-means clustering detected regional rainfall variability. Expert opinions were used to rank flood control strategies via the technique for order of preference via similarity to the ideal solution (TOPSIS) technique, with the best solutions emphasizing social, environmental, and economic factors. Taromideh et al. (2022) determined the likelihood of city floods without the use of conventional engineering models via ML approaches. The classification and regression tree (CART) algorithm performed the best, producing a very accurate flood hazard map for nations. River distance, volume of drainage, and vulnerable locations are important flood indicators. The technique provides effective outcomes in areas with little data when the AHP is combined with vulnerability.

This research proposes a novel, data-driven flood prediction model with the integration of big data analytics, socioeconomic factors, and risk management frameworks for more advanced flood prediction accuracy, which may enable the enhancement of resilience among more flood-vulnerable communities. This applies to the BO-EDNN for enhanced forecasting of flood events and its risk assessment.

The organization of the research is as follows: Part 2 presents the methodology and results, and the discussion is mentioned in Part 3, while Part 4 addresses the research's conclusions.

## 2. Methodology

This research proposes BO-EDNN, a novel model for flood prediction that combines big data analytics and social resilience frameworks to optimize parameters for better prediction results. The data utilized describe numerous elements crucial to flood prediction, such as environmental parameters and socioeconomic indicators. Data preprocessing, including cleaning and Z score normalization, is performed as an element of the data preparation process. PCA extracts crucial characteristics while lowering dimensionality. Figure 1 illustrates the comprehensive fundamental idea of the research flow diagram.

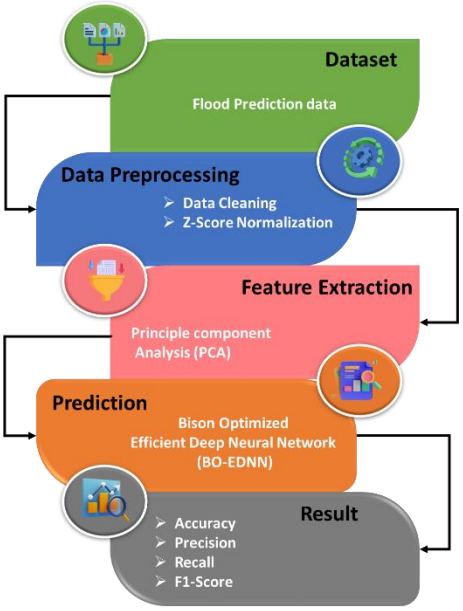



**Figure 1** Research overall fundamental concept diagram.

*2.1. Dataset*

Flood forecasting involves identifying, tracking, and warning individuals or organizations about the likelihood of a flood in a certain location. This approach involves the use of a variety of technologies and procedures to detect, anticipate, and reduce flood damage. Flood forecasting is an essential field of research because of the significant effects it has on structures, ecosystems, and human beings. Accurate flood prediction models can help with disaster preparedness and risk management, decreasing the negative impact of flooding. A variety of flood prediction features, such as environmental and socioeconomic indicators, were included. The database has fifty thousand rows and twenty-one columns. The data were collected from Kaggle: https://www.kaggle.com/datasets/naiyakhalid/flood-prediction-dataset.

*2.2. Data preprocessing*

Flood prediction using big data includes actions before the actual algorithm, such as data cleaning and Z score normalization, to increase the quality and consistency of the input variables. Data cleaning involves removing errors or irrelevant information, whereas Z score normalization involves standardizing the data so that every variable has the same amount as the predictive model. These steps improve the accuracy and certainty of flood risk management and social resilience activities.

Data cleaning aims to identify and fix incorrect data in a dataset. In this dataset, data cleaning checks for consistency in all numeric columns to obtain the correct type of int64 and checks for anomalies in value. Since there are no missing values, the main activities involve the removal of duplicates, correction during data entry, and verification that all features are within reasonable ranges.

Normalization is a big-data statistical methodology used to address the challenge of outliers in flood forecasting, as it tends to skew results. This transformation of flood-related features uses the standard deviation and mean, producing a normalized score with a mean of 0 and a standard deviation of 1, equation (1). This normalization helps produce a more accurate flood risk analysis, which can further advance social resilience and assist in risk management by making predictions more reliable.

$$w' = \frac{w\prime - \mu}{\sigma} \quad (1)$$

Where $\sigma$ represents the feature's standard deviation and where μ represents the feature's mean value. Values that are exactly similar to the mean are assigned to zero, values beyond the mean are shown as positive numbers, and values below the mean are shown as negative numbers when the Z score normalization approach is used. Once the data are cleaned

and normalized, the dataset should be divided into training and test sets. In this way, one can test the model's performance on unseen data. The data should be split via a standard 80--20 split, where both sets are representative of the overall dataset.

*2.3. Principal component analysis (PCA)*

When PCA is used for flood prediction and risk management, the aim is to obtain d-dimensional flood-related data in a space with fewer dimensions. This helps reduce space and time complexity and reduces the degrees of freedom when working with large datasets. If category labels were removed, each flood event in a dataset would become n-dimensional. The processes involved in the PCA calculation are shown in $w_1, w_2, \ldots, w_l \in q^n$. To capture key features linked to social resilience and flood risk prediction, equation (2) is used to find the mean vector μ in $m$ dimensions.

$$\mu = \frac{1}{l}\sum_{i=1}^{l} w_j \quad (2)$$

For the observed data, the calculated covariance matrix $T$ is shown in equation (3).

$$T = \frac{1}{l}\sum_{l=1}^{l}(w_j - \mu)(w_j - \mu)^s \quad (3)$$

Determine the latent values and matching latent vectors of $T$, where $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_k \geq 0$. The $l$ primary elements were determined from the $l$ initial variables via equation (4).

$$\begin{gathered} z_1 = b_{11}w_1 + b_{12}w_2 + \cdots + b_{1k}w_l \\ z_2 = b_{21}w_1 + b_{22}w_2 + \cdots + b_{2k}w_l \\ \ldots \\ z_l = b_{l1}w_1 + b_{l2}w_2 + \cdots + b_{kk}w_l \end{gathered} \quad (4)$$

$Z_l$must be unrelated. The initial variation in the dataset is explained to the greatest extent feasible by $z_1$, the remainder of the variance is explained to the greatest extent possible by $z_2$, etc. The most useful datasets often have a few numbers of larger latent values that predominate over the rest of equation (5),

$$\gamma_l = \frac{\lambda_1+\lambda_2+\cdots+\lambda_n}{\lambda_1+\lambda_2+\cdots+\lambda_n+\cdots+\lambda_l} \geq 80\% \quad (5)$$

Where the proportion kept in the data representation is denoted by $\gamma_l$. The primary components that can account for at least 80% of the total variance should be retained when PCA is employed for feature extraction.

*2.4. Bison-optimized efficient deep neural network (BO-EDNN)*

The BO-EDNN model is utilized to improve the accuracy of flood prediction by using a bison-inspired optimization algorithm to tune the network parameters. The BO-EDNN model combines big data analytics with social resilience factors such as community preparedness, adaptive capacity, and infrastructure vulnerability for a holistic flood prediction system. This approach combines traditional hydrological and meteorological data with social and infrastructural insights to enhance a more holistic understanding of flood risk. With an optimized deep neural network, the model enhances the predictive performance and can be used as a strong tool for flood forecasting and risk management. It incorporates social resilience aspects such that it includes human and infrastructural responses to flood events; therefore, the model provides enhanced overall risk assessment and preparedness strategies for vulnerable communities.

*2.4.1. Efficient deep neural network (EDNN)*

*The* EDNN increases the predictive capacity for handling complex and high-dimensional flood data. The ability of the EDNN to capture complex patterns and relationships present in large datasets further enhances the accuracy and reliability of flood predictions. Figure 2 illustrates an EDNN with 3 layers: a hidden layer, a layer of output, and a layer of input. Each layer consists of many nodes that are gradually connected to form a complete node in the next layer. Usually, both the input and output layers are one, whereas hidden layers might have multiple layers. The suggested approach that uses the input data is analyzed using 6 layers of input and 7 layers of hidden data. The EDNN has sixty-four neurons as input and sixty-four neurons as output for each fast current cycle. The majority of neurons are overfitting, leaving only a few underfitting. The size and quantity of the hidden layer neurons are carefully selected for the aforementioned reasons. Each neural layer is computed in practice. The TensorFlow executes the size layer of the hidden space, which is configured using hyperparameters. The problem-solving and fault-learning abilities of an EDNN rely on its activation function. The input layer of the EDNN receives the coefficient value, whereas the output layer receives the predicted classes. Every node's weight is computed and forecasted via a function of activation. The suggested method employs a rectified linear unit (ReLU) as a function of activation to optimize node weights and minimize system errors. Pack propulsion is used to move the weight from the layer of the output back to the layer of the input, reducing the cost function. Neuronal outputs are defined as follows:

$$z_r^{m+1} = \sigma(y) = \sigma\left(\sum_{j=1}^{n} x_{jr}^m z_j^m + a_r^{m+1}\right) \quad (6)$$

In equation (6), $\sigma(y)$ represents the activation function, $z_r^{m+1}$ represents the output of $r$ neurons in the $m+1$ layer, $x_{jr}^m$ represents the weights across the $m$ layer's $j$ neurons and the $m+1$ layer's $r$ neuron, and $a_r^{m+1}$ represents linear relationship biases. The function of loss measures the differences across the computed and actual coefficient values during the repulsion process. The factors $a$ and $\omega$ were fixed by a small function of the loss value. The function of loss to the enhanced DNN is expressed as follows in equation (7):

$$F(\theta) = \frac{1}{M}\sum_m \sum_r s_{mr} log z_{mr} \quad (7)$$

Where $s_{mr}$ is the true value of the $r^{th}$ specimen $n^{th}$ component, $z_{mr}$ is the anticipated value of the $r^{th}$ sample $m^{th}$ element, $\theta$ represents the parameters $\omega$ and $a$, and $M$ is the integer of the sample. The implementation of a dropout system minimizes neuron outfit, upsetting the neural network topology by eliminating neurons. Furthermore, the proposed strategy enhances the conventional learning rate over the usual gradient-descending system. The optimum amount of $\theta$ is represented as follows (equations (8 and 9):

$$\{n_s = \beta_1 n_{s-1} + (1-\beta_1)h_s \; U_s = \beta_2 v_{s-1} + (1-\beta_2)h_s^2 \; h_s = \nabla_\theta F(\theta_{s-1}) \; \hat{n}_s = \frac{n_s}{1-\beta_1^s} \; \widehat{U}_s = \frac{v_s}{1-\beta_2^s} \; \theta_s = \theta_{s-1} - \alpha \frac{\hat{n}_s}{\sqrt{\hat{u}_s + \varepsilon}} \quad (8)$$

$$\alpha = \alpha_0 \beta_3^{\frac{\frac{epoch-num}{M}}{batch-size}} \quad (9)$$

In this equation, $U_s$ represents the mean motion of the square of the gradient, $h_s$ represents the value of the parameter gradient, $n_s$ represents the average motion of the gradient, $\alpha_0$ represents the learning rate of the beginning value, $\widehat{U}_s$ and $\hat{n}_s$ are corrected quantities, $\beta_1, \beta_2, and\ \beta_3$ represents the exponential decay rates $0.9, 0.999, and\ 0.95$, the computation group parameter is $batch - size$, and the training time is $epoch - num$.

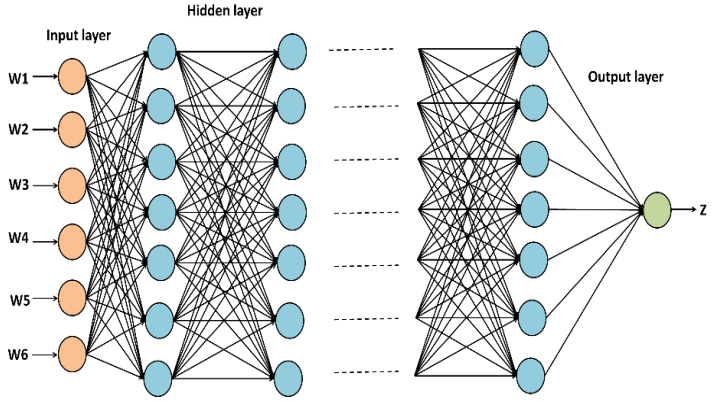



**Figure 2** Architecture of the EDNN.

*2.4.2. Bison optimization (BO)*

*The* BO algorithm was developed to improve the effectiveness of the flood prediction model along with its accuracy. By utilizing the natural behavior of bison, the BO algorithm has explored the problem to optimize its parameters for a greater number of EDNNs. The outcome of the above is improved model performance, yielding very effective predictions as well as risk management capabilities, especially in the management of complex, multidimensional data. The BO was modeled after typical bison patterns of behavior. Bison has 2 unique defense mechanisms: creating a circle around weaker individuals and running inexhaustibly. Two groups define the algorithm, each of which simulates a distinct behavior. The swarming pattern is modeled by the first group. The first step is to determine the center of the fittest swarming individuals, who are arranged according to the value of the goal function. The ranked center calculation is used. This method calculates the center for each individual's weight, as shown in equation (11), after sorting the fittest individuals and assigning appropriate weights on the basis of the quality of the solutions represented in equation (10). Only when a new position candidate increases the swarmer's objective value function is it employed in equation (12).

$$weight = (10,20,\ldots,10 * t) \quad (10)$$

$$D = \sum_{j=1}^{T} \frac{weight_j * w_j}{\sum_{i=1}^{t} weight_j} \quad (11)$$

$$w_{new} = w_{old} + (d - w_{old}), rand(0, v) \quad (12)$$

Here, $t$ denotes the elegant size of the group, defines the integer of fittest bison for determining the center, $D$ represents the center, $\upsilon$ constitutes the exceedable variable, $w_j$ represents the ith answer, the present solution is $w_{old}$, whereas the new solution contender is $w_{new}$. The second group replicates running behavior. The method generates a randomized run position vector ($q$) during initiation in equation (13) and adjusts it slightly after each iteration, as shown in equation (14). The running movement follows the run position vector described in equation (15), regardless of the solution quality. Unlike swarmers, runners move whenever quality is threatened.

$$q = rand\left(\frac{ub-lb}{45}, \frac{ub-lb}{15}\right) \quad (13)$$

$$q = q.rand(0.9,1.1) \quad (14)$$

$$w_{new} = w_{old} + q \quad (15)$$

The search space boundaries are indicated by $ub\ and\ lb$, the run position vector by $q$, and the original and new solutions by $w_{old} and w_{new}$. The Bison algorithm interprets the search space as a hypersphere, allowing for direction-based accessibility throughout the optimization process. Individuals who cross boundaries explore the whole search space. Table 1 describes the control variables for the BO. The suggested value of 3.5 for the exceedance parameter allows the swarmer to overshoot the center 2.5 times, according to equation (12). Equation (16) defines the running size of groups on the basis of the swarm size of groups.

$$Run\ group\ size = MO - swarm\ size \quad (16)$$

The number of populations ($MO$) and swarming group size (swarm size) are the parameters used.

**Table 1** BO control variables.

| Variables | Description |
|---|---|
| Number of population (MO) | Size of population |
| Elegant size of the group | The integer of best-fitting solutions utilized to calculate the center |
| The swarm size of the group | Integer of swarm group solutions' |
| υ Exceed variable | Highest swarming motion length relative to the center. 0 indicates zero motion. 1 corresponds to the center. Recommended range: 3.5-4.1. |

## 3. Results

Python 3.6.3 was used as the software language, with a CORE I7 CPU, 64 GB of RAM, a 256 GB solid-state operation, a 1070 GPU, and the Windows 10 operating system. The influence of key factors of social resilience on flood prediction and risk management in the proposed BO-EDNN model is shown in Figure 3. It depicts how each factor, that is, community preparedness, adaptive capacity, and infrastructure vulnerability, contributes to the performance of the model. The flood prediction score represents how each factor helps the model predict flood events, whereas the risk management score represents the impact of each factor in managing and mitigating flood risk. The scores of these models show how these various socioeconomic factors are combined in an integrated flood prediction and risk management approach within the model.

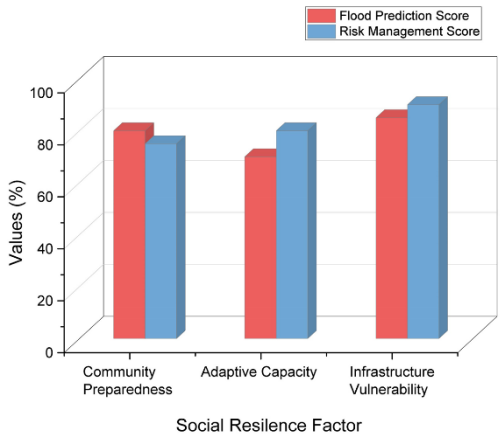



**Figure 3** Social resilience for flood prediction and risk management.

The outcomes show how well the BO-EDNN predicts floods in comparison with other conventional techniques. The conventional techniques include SVM (Zhang et al., 2023) and RF (Zhang et al., 2023). The performance was subsequently evaluated in terms of several metrics, including the recall, accuracy, precision, and F1 score.

The fraction of actual positives or real negatives that are accurate forecasts relative to the total is known as accuracy. Figure 4 illustrates the accuracy comparison outcome. This reflects that the BO-EDNN model can achieve better (95.2%) performance than traditional methods (SVM -79.2% and RF-88.6%) since it has more information about flood risk with diverse data sources. Better predictions are anticipated from the BO-EDNN model's higher probability of learning more types of influencing elements that affect flood occurrence.

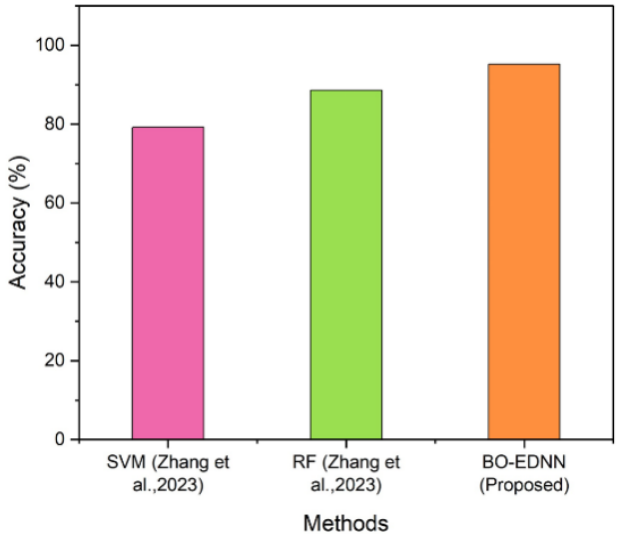



**Figure 4** Comparison of accuracy.

Precision is considered the ratio of true positives, that is, correctly predicted flood events, to overall positive predictions proposed by the model. Figure 5 illustrates the precision comparison outcome. Since BO-EDNN (96.1%) integrates more factors, it can make more reliable flood predictions, reducing the possibility of false positives and hence improving precision. Conventional models (SVM -83.5% and RF-94.9%), which are frequently limited to climatic or hydrological data, can underestimate floods in locations where they are less likely to occur, resulting in decreased precision due to more false positives.

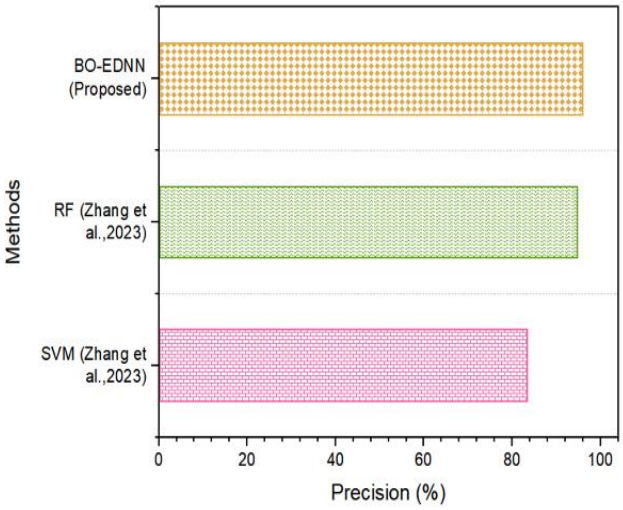



**Figure 5** Comparison of precision.

Recall refers to the proportion of real flood events correctly identified as floods by the model. Figure 6 illustrates the recall comparison outcome. BO-EDNN (92.6%) performs better than other traditional models (SVM -83.9% and RF-86.9%). In terms of recall, it can detect more actual flood events, which are essential for flood prediction and early warning. Adding several factors, such as historical data on flooding and socioeconomic contexts, has made the BO-EDNN more capable of identifying real flood events.

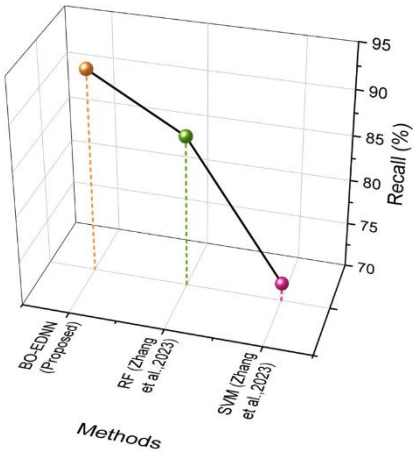



**Figure 6** Comparison of Recall.

The F1 score offers a balance across recall and accuracy by taking the harmonic mean of the two. Figure 7 illustrates the F1 score comparison outcome. An imbalance between recall and accuracy in standard models (SVM -83.7% and RF-90.7%) might result in an inferior F1 score than what could be obtained with more accurate data and optimization. BO-EDNN (93.2%) improves both the recall and accuracy in a balanced way, thus providing a higher F1 score. This means that the BO-EDNN has better performance in maintaining a good balance between the detection of floods and avoiding false predictions. Table 2 shows a comparison of various methods.

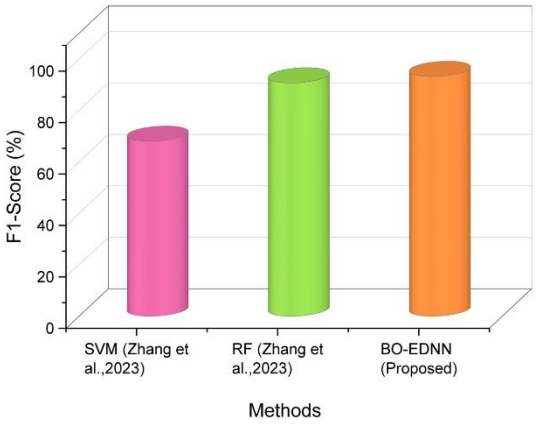



**Figure 7** Comparison of F1 scores.

**Table 2** Numerical outcomes of performance evaluation based on various techniques.

| Techniques | Accuracy (%) | Precision (%) | Recall (%) | F1 score (%) |
|---|---|---|---|---|
| SVM (Zhang et al., 2023) | 79.2 | 83.5 | 83.9 | 83.7 |
| RF (Zhang et al., 2023) | 88.6 | 94.9 | 86.9 | 90.7 |
| BO-EDNN (Proposed) | 95.2 | 96.1 | 92.6 | 93.2 |

## 4. Discussion

This research highlights the importance of combining socioeconomic factors in flood prediction models, which are overlooked in traditional approaches. Nakhaei et al. (2023) introduced a smart flash flood risk assessment framework but emphasized dry climates, which restricts its applicability to various geographical settings. The method allows the use of qualitative indicators of resilience, which are not necessarily accurately quantifiable for machine learning applications. The framework also does not utilize sophisticated deep learning methods to adapt dynamically to changing flood patterns. This research prioritizes community resilience without incorporating real-time sources of data for adaptive flood forecasting.

Dong et al. (2022) applied deep learning coupled with multiwatershed flood monitoring but relied mainly on physical and social sensor data, possibly lacking important socioeconomic factors such as income, education, and infrastructure vulnerability. High-quality sensor networks are needed in their method, which might not be accessible everywhere, thus proving challenging to adopt in resource-scarce settings. It uses no dimension reduction methods, creating data redundancy issues and computational intensity. It also fails to respond to how adaptively deep learning models can re-evaluate forecasted predictions within long-term flood resilience planning.

Ghaith et al. (2022) proposed a synchronization-augmented deep learning model for early flood risk forecasting, but their model centers almost exclusively around urban flood conditions and does not consider rural or blended environments where flood processes are much different. Their model does not adequately incorporate socioeconomic variables such as community readiness and risk perception, which are important in disaster response. Although their deep learning model improves real-time predictions, it does not have an optimization process to adaptively modify learning parameters for enhanced accuracy. Their approach also does not utilize feature extraction methods such as PCA to enhance computational effectiveness and model explainability. Zhang et al. (2023) proposed a machine learning-based urban flood risk assessment model, but their work mainly addresses static flood risk indicators instead of dynamic predictors in real time. Their method lacks a deep learning-based optimization mechanism, which restricts its ability to identify nonlinear patterns in floods over time. Finally, their use of traditional machine learning methods could restrict their model scalability and flexibility to other flood situations.

The proposed BO-EDNN model surpasses current limitations by combining hydrological, meteorological, and socioeconomic factors into a single predictive model, rendering it versatile across varying environmental conditions. It uses principal component analysis (PCA) to reduce dimensions, providing effective feature extraction and enhanced computational efficiency. In contrast to traditional approaches that are dependent only on sensor readings, the proposed method uses big data analytics and social resilience metrics for improved robustness even in sparse data situations. The model also employs metaheuristic optimization (Bison optimization algorithm) for dynamically adjusting to changing flood patterns, resulting in improved accuracy. Additionally, the adaptive mechanism of the deep learning-based architecture provides real-time response as well as scalability for both urban and rural flood-exposed areas.

## 5. Conclusions

Research aims to develop a new data-driven flood prediction model that employs big data analytics through a combined framework of social resilience and risk management. The proposed model is expected to generate highly accurate flood event predictions while explicitly considering the socioeconomic impact on vulnerable populations. It proposes the so-called BO-EDNN as the core model for prediction. This model theoretically enhances flood prediction by optimizing network parameters via the bison-inspired optimization algorithm. The dataset used in this research encompassed several relevant features in flood prediction, including environmental factors and socioeconomic indicators. Data cleaning involves eliminating missing values and abnormalities to remove variations. Z score normalization scales the data and ensures balance for all input variables, which helps improve convergence and data quality. Focusing on feature extraction, PCA aims to perform dimensionality reduction to retain key variance in the dataset used. The results indicate that the proposed BO-EDNN model displays superior efficacy to traditional flood prediction methods. The model performance was evaluated in terms of prediction accuracy (95.2%), precision (96.1%), recall (92.6%), and F1 score (93.2%). A model for flood prediction that integrates big data analytics with social resilience frameworks has been successfully developed. The BO-EDNN model, which has improved accuracy in predicting floods, also addresses aspects concerning socioeconomics for sustainable and resilient flood management systems.

### *5.1. Limitations and future directions*

Flood prediction using big data for social resilience and risk management is constrained by the quality of the data, the data integration issue, and the computational ability. Specifically, prediction model performance can be highly influenced by real-world data quality and local contextual factors. The future direction lies in further enhancing the integration of data and opening access to community-based knowledge to facilitate more adaptive and precise flood risk management.

## Ethical Considerations

Not applicable.

## Conflict of Interest

The authors declare no conflicts of interest.

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